

Transfer Attacks in Face Recognition

Chirag Sharma

June 14, 2026

Baseline Attacks (already implemented):

- PGD (baseline)
- MI-FGSM (momentum-based)
- TI-FGSM (translation-invariant)
- SI-NI-FGSM (scale-invariant + Nesterov)
- MI-ADMIX-DI-TI (combined augmentation)

New Attack Implemented:

- **BSR - Block Shuffle and Rotation (CVPR 2024)**

BSR — Block Shuffle and Rotation

Definition & Nature

- Input-transformation based transfer attack
- CNN-targeted, no ViT components needed
- Published at CVPR 2024
- Paper: [arXiv:2308.10299](https://arxiv.org/abs/2308.10299)

Key Insight

- Existing attacks overfit to surrogate model's spatial attention heatmap
- Breaking spatial structure disrupts model-specific attention
- Gradient averaged over many shuffled views lands in a flat, transferable region

Why BSR Fits Face Recognition

- All surrogate models are CNNs:
FaceNet512, ArcFace, GhostFaceNet, VGG-Face
- CNN attention is spatially localised - shuffling breaks it effectively
- Forces perturbation into the global, model-agnostic embedding space
- ViT-based attacks (ATT) failed here because ViT attention is token-based, not spatial

BSR Transform (one copy)

- 1 Split image along **width** into `num_block` strips of random size
- 2 **Permute** the width strips randomly
- 3 For each width strip, split along **height** into `num_block` strips and permute
- 4 **Rotate** each cell by a small random angle ($\theta \sim \mathcal{N}(0, 0.05)$ rad)
- 5 Reassemble the image

Repeat `num_copies` times, average all gradients, then do MI-FGSM

Hyperparameters

Parameter	Value
<code>num_copies</code>	20
<code>num_block</code>	2
rotation stddev	0.05 rad
ϵ	0.062
iterations	5
decay μ	1.0

BSR Goal in Face Recognition

- Operates in embedding space
- Impersonation: increase cosine similarity
- Dodging: decrease cosine similarity

Attack Implementation

BSR in Our Implementation

- 1 `_bsr_get_lengths()` samples random strip sizes from $\mathcal{U}(0, 1)$, rounds, and ensures exact sum
- 2 `_bsr_shuffle_rotate()` splits W strips $\rightarrow$ permutes $\rightarrow$ for each W strip splits H strips $\rightarrow$ permutes $\rightarrow$ rotates
- 3 `bsr_attack()` uses `tf.Variable` for `adv` so `GradientTape` tracks gradients *through* the transform back to the perturbation
- 4 Rotation via `tf.raw_ops.ImageProjectiveTransformV3` (pure TF2, no external packages)
- 5 Cells smaller than 4 px skip rotation to avoid degenerate splits

Critical Bug Fixed

- Naive port used `tape.watch(x_t)` on the *transformed* copy — gradient never reached `adv`
- Fix: `adv` is a `tf.Variable`; `tape` watches it automatically through the transform
- This single fix raised breach rate from **11.46%** to **36.67%**

Resources

- Official paper: [arXiv:2308.10299](https://arxiv.org/abs/2308.10299)
- Our implementation: github.com/chiraagsharma2

Results — Overall Comparison

All Attacks on 30-Pair Subset

Attack	Breach %	Impact
BSR (ours)	36.67	0.1969
SI_NI_FGSM	31.46	0.1788
MI_FGSM	26.04	0.1618
MI_ADMIX_DI_TI	23.96	0.1509
TI_FGSM	20.00	0.1290
PGD	16.46	0.1011

baseline: 5 vanilla attacks

Key Results

- BSR is **#1** — beats all 5 vanilla baselines
- **+5.21%** over best baseline (SI_NI_FGSM at 31.46%)
- Impact mean 0.1969 vs 0.1788 — stronger embedding-level effect

Results — By Goal

Dodging Attack

Attack	Breach %
BSR	48.75
SI_NI_FGSM	38.75
MI_FGSM	33.75
MI_ADMIX_DI_TI	30.42
TI_FGSM	26.67
PGD	22.08

Impersonation Attack

Attack	Breach %
BSR	24.58
SI_NI_FGSM	24.17
MI_FGSM	18.33
MI_ADMIX_DI_TI	17.50
TI_FGSM	13.33
PGD	10.83

- BSR is substantially stronger on **dodging** (48.75%) — spatial disruption makes it easy to fall below the verification threshold
- BSR also leads **impersonation** (24.58%) — crossing into a different identity's embedding region is harder but BSR still outperforms all baselines

Results — Attacker → Victim Breakdown (BSR)

BSR Breach Rate per Attacker

Attacker	Victim	Breach (%)
Facenet512	ArcFace	66.67
VGG-Face	Facenet512	66.67
Facenet512	VGG-Face	63.33
VGG-Face	ArcFace	60.00
Facenet512	IR152	46.67
ArcFace	GhostFaceNet	43.33
Facenet512	GhostFaceNet	43.33
VGG-Face	GhostFaceNet	43.33
ArcFace	Facenet512	36.67

Observations

- **VGG-Face** and **FaceNet512** are the strongest surrogate models for BSR
- **IR152** is the hardest victim — BSR breach rates are lowest here, consistent with all other attacks
- **GhostFaceNet** as attacker transfers least effectively — its compact architecture produces less generalisable gradients
- Cross-architecture pairs (e.g. VGG-Face → ArcFace) show high breach rates, confirming BSR's strong cross-model transferability

Thank You

Paper	arXiv:2308.10299
Official repo	Trustworthy-AI-Group/BSR
Our implementation	chiraagsharma24/BSR-Implementation